

Kexin Wei

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Robotics & medical-devices engineer specializing in motion control, surgical imaging, and sensor integration, with a track record of clinically validated, animal-trial-tested systems.

Experience

Research Engineer <i>Imperial Global Singapore.</i>	Mar 2025 — Present <i>Singapore</i>
Real-Time Control: Implemented and tuned a real-time closed-loop glucose-insulin patient controller (MATLAB → Python) within a multi-container simulation stack. Systems: Built a Flask + Vue.js application for real-time simulation and cyberattack visualization. Security: Researching agentic automated penetration testing for medical cyber-physical systems. Infrastructure: Dockerized deployment and administers lab workstations, servers, and container infrastructure.	
Software Team Lead <i>Creative Medtech Solutions Pte.Ltd.</i>	Mar 2024 — Jan 2025 <i>Singapore</i>
Robot: Engineered robotics control systems including kinematics optimization for 4-DoF robot and control interfaces. Image: Developed 4D DICOM processing and US/MRI registration algorithms for medical image alignment. CyberSecurity: Enhanced security through encryption protocols, SQL injection protection, and streamlined Qt based GUI. Lead: Led a team of 6 engineers: Git workflow, Scrum adoption, GoogleTest integration, and code-review standards.	
Research and Development Engineer <i>Creative Medtech Solutions Pte.Ltd.</i>	Jun 2022 — Mar 2024 <i>Singapore</i>
Robot: Designed a motion-tracking algorithm for a 6-DoF robot arm integrating an A*STAR CNN, verified in vivo across pig and dog trials. Sensor: Integrated electromagnetic sensor system in the HIFU treatment software for 3D visualization of the ultrasound probe. Signal Processing: Created HIFU power-calibration and kidney respiratory-tracking algorithms, verified in ex-vivo and animal trials. Patent: Filed 3 patents in medical algorithm design: HIFU calibration, respiratory tracking, and time synchronization.	

Education

M.Eng in Medical Robotics and AI <i>National University of Singapore, Singapore</i>	Aug 2020 — May 2022 <i>GPA: 4.63/5.00</i>
- Thesis: Deep Reinforcement Learning (DRL) for Robot-assisted Surgical Training. Code. - TA for ME2143E Motor Characteristics and ME5405 Machine Vision.	
B.Eng in Mechanical Engineering <i>Tongji University, China</i>	Sep 2014 — Jul 2019 <i>GPA: 4.61/5.00</i>

Awards & Honors

AI Mathematics Modeling Scholarship	Hackathon “Runner Up” Prize at bio LLMs hosted by 4Catalyzer (2023) 2nd Prize in CUMCM (2018); Honorable Mention in MCM/ICM (2018) 3rd Class Award (2018); 1st Class Award (2016); 3rd Class Award (2015)
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Skills

Robotics Programming	ROS, 6-DoF motion control, kinematics, EM/US tracking, VTK, OpenCV, DICOM
AI/ML	C++, Python, MATLAB, Qt, Linux, Git, Docker, SQL, Flask, Vue.js
CAD/CAE	PyTorch, TensorFlow, Deep Learning (CV, CNN, DRL), segmentation (SAM/MedSAM)
Language	Solidworks, AutoCAD English (C1), German (C1), Mandarin (native)